Motor Imagery EEG Classification with Real-World and EEG-GAN-Generated Data

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***Abstract*—Accurate classification of EEG signals is crucial for applications such as brain-computer interfaces (BCI). In this study, we examined the effectiveness of long short-term memory (LSTM) networks for classifying EEG signals recorded during six different motor imagery tasks. The dataset included recordings from 16 electrodes and consisted of actions such as forearm supination, elbow flexion, forearm pronation, hand closing, hand opening, and elbow extension. Our model achieved an average F1 score of 0.87 on the real-world EEG data. To evaluate the feasibility of training the model on similar synthetic data, we generated synthetic EEG signals for hand opening and closing tasks using EEG-GAN. When tested on synthetic data, the LSTM model trained solely on real data achieved the average F1 score of 0.76. These results highlight the potential of generative models for augmenting EEG datasets and the robustness of LSTM-based classifiers.**

***Index Terms*—EEG, LSTM, EEG-GAN, generative models, BCI.**

I. Introduction

Electroencephalography (EEG) is a widely used non-invasive method for recording brain activity applied in medicine, neuroscience, and brain-computer interfaces (BCI). However, classifying EEG signals remains challenging due to their high dimensionality and variability. Recent advances in deep learning, such as recurrent neural networks (RNNs), particularly long short-term memory (LSTM; [1]) networks, have demonstrated high efficiency in modeling the temporal structure of EEG data. Additionally, generative adversarial networks (GANs; [2]) open new opportunities for creating synthetic data, which can help address the problem of scarce labeled data for training models.

This study aims to evaluate the performance of an LSTM model in classifying EEG signals corresponding to six hand-related motor imagery tasks. Furthermore, we investigate the use of EEG-GAN [3] for generating synthetic signals.

II. Related work

EEG signal classification involves assigning each temporal segment of recorded signals to specific categories corresponding to cognitive states, motor imagery, pathological conditions, or other parameters. This task plays a crucial role in various fields, such as medical diagnostics, neurophysiological research, and BCI development [4].

Traditional EEG classification methods typically involve manual feature extraction using techniques such as Fast Fourier Transform (FFT) or wavelet transform [5], followed by classification using conventional machine learning algorithms, including Support Vector Machine (SVM) [6], [7] or Random Forest [7]. These approaches have been successfully applied to various EEG-based tasks, such as seizure detection [8] and cognitive workload estimation [9]. However, these traditional approaches require manual feature selection [10], which is necessary to reduce dimensionality and improve classification accuracy. Nonetheless, traditional feature selection methods may not fully optimize feature subsets, potentially increasing computational complexity.

Recently, deep learning methods have demonstrated significant improvements in EEG classification. Convolutional Neural Networks (CNNs) effectively extract spatial patterns from multi-channel EEG recordings, successfully distinguishing between specific cognitive states or conditions in motor imagery tasks [11]. CNNs have been widely applied in motor imagery classification for BCI systems [12], [13] and emotion recognition [14]. Despite their strengths, CNNs do not sufficiently capture long-term temporal dependencies, which are crucial for accurate EEG signal interpretation. RNNs, especially LSTM networks, address this limitation by efficiently modeling sequential data and retaining temporal dependencies over extended periods, thus making them more suitable for EEG analysis tasks. Studies have shown that LSTM-based models outperform traditional machine learning classifiers in tasks such as epileptic seizure detection [15] and sleep stage classification [16], [17].

A major challenge in EEG classification is the limited availability of sufficiently large annotated EEG datasets. Acquisition of real-world EEG data is often constrained by a number of factors, including strict confidentiality and data protection requirements, ethical considerations [18], and the need for specialized equipment and qualified personnel [4]. These constraints limit data availability, negatively impacting the scalability and generalizability of EEG research. Additionally, studies have highlighted high variability in EEG recordings due to inter-subject differences and environmental factors, making it difficult to standardize datasets [19].

A promising approach to overcome these limitations is generating synthetic EEG data using GANs. GANs enable the creation of synthetic EEG signals statistically and structurally similar to real-world EEG data, which makes it possible to expand training datasets without collecting additional real-world data. Therefore, the use of synthetic data can improve performance and generalization capabilities of classifiers: thus, previous research has demonstrated that GAN-generated EEG signals improved accuracy of seizure detection [20]. Furthermore, synthetic data do not face the ethical, legal, and privacy restrictions that typically accompany real EEG data collection, which makes them highly advantageous for research purposes [21] [22].

To sum up, the integration of deep learning methods, particularly LSTM and GAN-based synthetic data generation, represents a promising research direction for enhancing EEG classification performance, addressing data availability constraints, and advancing BCI technology development.

III. Methods

A. Real-World Dataset

For training the model, we used an open-access dataset [23], which provides EEG recordings collected during motor imagery tasks. The dataset was acquired as a part of a recent study [24]; in the following, we report the relevant aspects of data acquisition and preprocessing.

The dataset contains EEG recordings collected during an experiment involving imagining six different movements of the right upper limb:

1) Elbow extension
2) Elbow flexion
3) Forearm pronation
4) Forearm supination
5) Hand closing
6) Hand opening.

The experiment involved 10 healthy participants (5 males and 5 females) aged between 24 and 38 years (mean age: 30 years). All participants were right-handed. During the experiment, participants sat in a comfortable chair facing a computer screen that displayed visual stimuli instructing them to perform specific motor imagery tasks. The right arm was placed naturally on a table in a semi-flexed position at a 120-degree angle to minimize muscle fatigue and maintain stable EEG signals.

The experimental paradigm was cue-based, with visual instructions displayed on the screen in a structured sequence. Each participant completed 10 sessions, each consisting of 180 trials (30 trials per movement class). Each trial lasted 6 seconds and followed this sequence:

- 0 sec: A fixation cross appeared on the screen, requiring the participant to focus.
- 2 sec: A visual cue appeared, indicating which movement to imagine.
- 4 sec: A rest period before the next trial began.

Each participant completed 300 trials per movement class, ensuring the acquisition of robust data for motor imagery analysis.

EEG signals were recorded using 16 active Ag/AgCl electrodes connected to a 16-channel OpenBCI CytonDaisy amplifier. The electrodes were placed according to the 10–20 international system:

- Reference electrode: placed on the left earlobe.
- Ground electrode: placed on the right earlobe.

The EEG signals were sampled at 500 Hz and initially filtered using an 8th-order Chebyshev bandpass filter (0.01–200 Hz), with additional suppression of power line interference via a 50 Hz notch filter. Data acquisition was performed using the *OpenBCI GUI* software, and the resulting dataset, provided in `.set` format, is organized by participant, session, and trial, with event codes denoting the corresponding movement class.

Offline preprocessing was carried out in *MATLAB R2020b* using the *EEGLAB* toolbox to improve signal quality and remove artifacts. This included bandpass filtering (0.01–200 Hz), artifact removal through Independent Component Analysis (ICA) to eliminate muscle activity, eye movement, and line noise, as well as manual inspection of all trials to discard segments with abnormal patterns, extreme values, or excessive head motion. The preprocessing ensured that the dataset is well-curated and standardized, making it a valuable resource for research on motor imagery, brain-computer interfaces, and neural signal analysis.

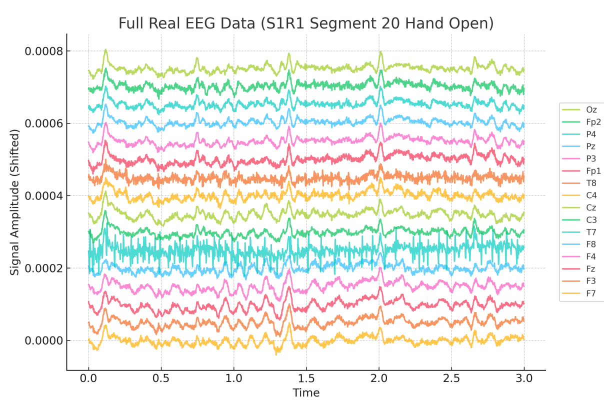

Fig. 1. Example of EEG signal during hand opening, hand opening signal no. 20. Amplitude has been shifted for better visualisation.

B. Data Generation with EEG-GAN

Synthetic data were generated using the EEG-GAN approach [3] as implemented in [25], utilizing GANs specifically designed for EEG signal augmentation. The process consisted of the following steps:

1) *Autoencoder (AE) Training*. The autoencoder was trained to encode and decode single-channel EEG segments, aiming to create a compact representation of the original signals. The AE was trained for 2000 epochs on individual electrode segments (e.g., segment 0–100, segment 100–200, etc.). Due to the structured nature of the EEG signals, the autoencoder achieved minimal reconstruction loss of approximately 2e-5.
2) *GAN Training*. The primary challenge in generating EEG signals was determining the optimal segment length (i.e., the number of datapoints), as this parameter influenced not only the training duration for both the AE and GAN but also the final classification process. The real-world dataset consisted of EEG signals with a maximum possible length of 1500 datapoints. However, upon analyzing the data, it was evident that the signals contained distinct patterns in certain regions. This allowed for segmentation into shorter sequences, ensuring that each segment preserved meaningful patterns for training while significantly reducing computational demands.
   For the training dataset, signals were divided into segments of 100 time units per electrode, all stored in a single CSV file. To simplify the classification task, we opted to generate only two types of signals — open hand and closed hand movements — as they represent closely related motor actions that a neural network should be able to distinguish with high accuracy. Consequently, the training dataset for the GAN consisted of 282 open-hand movement signals and 286 closed-hand movement signals, each partitioned into multiple 100-time-unit segments.
   Unlike the autoencoder, which successfully learned single-channel representations, training the GAN on individual electrodes proved ineffective. When trained on a single-channel dataset, the generator and discriminator reached loss values of approximately 9.4 and 8.1, respectively. However, visual inspection of the generated signals revealed that the output lacked the characteristic patterns of real EEG signals, instead appearing chaotic. This issue arose because the generative model failed to capture the multichannel dependencies inherent in EEG data. To address this, we trained the GAN simultaneously on all 16 electrode channels, ensuring that the generated signals retained the spatial and temporal correlations present in real EEG data. The GAN training process required substantial computational resources: training the generator for 1200 epochs with a batch size of 128 and a learning rate of 1e-4 took approximately 16 hours. Since the model was trained on 100-time-unit segments, multiple GAN instances were used to generate longer, realistic EEG signals in a segmented manner. The final full-length EEG signals were reconstructed by concatenating these generated segments, preserving the temporal coherence of the data.
3) *Synthetic Data Generation*. Using the trained generator, synthetic EEG signals were generated for all 16 electrodes, ensuring that the synthesized data closely resembled real EEG activity while maintaining the key statistical and structural properties required for further classification. An example of a generated EEG signal assembled from segments each 100 time units in length is provided in Fig. 2.
   The GAN model was trained on an NVIDIA A100 GPU with 40GB of VRAM. The training process required the installation of the EEG-GAN library [26], which depends on PyTorch 1.12.1 and automatically resolves all additional dependencies, including NumPy and Pandas. Signal generation was also done on this setup.

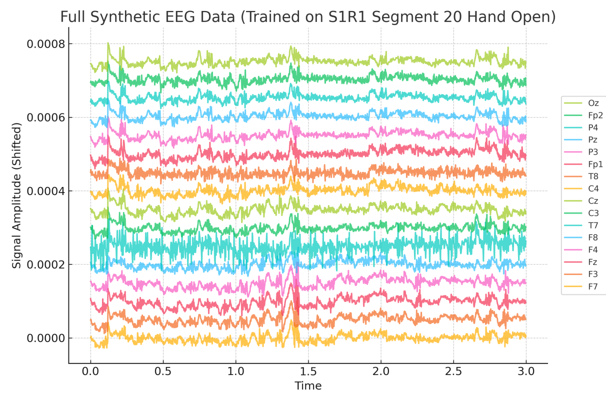

Fig. 2. Example of generated EEG signal during hand opening. Generation based on real hand opening signal no. 20. Amplitude has been shifted for better visualisation.

C. LSTM for classification task

We developed an EEG classification model based on a LSTM architecture proposed in [27]. The input dataset consisted of 2159 EEG files that were pre-segmented into 100-time-point fragments. After segmentation, the training set contained 16185 segments distributed across six classes as follows:

- elbow extension (class 0): 2700 segments;
- elbow flexion (class 1): 2700 segments;
- forearm pronation (class 2): 2700 segments;
- forearm supination (class 3): 2700 segments;
- hand close (class 4) : 2700 segments;
- hand open (class 5): 2685 segments.

The data were split into 80% for training, 10% for validation, and 10% for testing the model. Therefore, the test set consisting of 108 files was used for the final model evaluation.

Before being fed into the neural network, the data underwent several preprocessing steps. First, all signals were normalized using MinMax scaling, bringing the values into the [0,1] range to accelerate model convergence. Next, for each segment, the target label was assigned based on the most frequently occurring class within that time window.

The model architecture (see Fig. 3) consisted of two sequential LSTM layers with a hidden state dimension of 256. Batch normalization was applied before the first layer to stabilize the input data distribution, while a dropout mechanism (with a rate of 0.3) was used after each LSTM layer to reduce overfitting. The final fully connected layer converted the hidden representation into a probability distribution over the six classes. The model was trained using a cross-entropy loss function with label smoothing (0.1), and the Adam optimizer was employed with an initial learning rate of 0.001. The training was conducted in mini-batches of size 128 for a total of 1000 epochs, and a step-based learning rate scheduler was used to reduce the learning rate by a factor of two if the validation accuracy did not improve for 200 consecutive epochs.

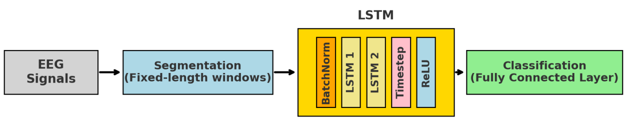

Fig. 3. Architecture of the LSTM-based neural network for EEG signal classification.

The LSTM classification model was trained on RTX3070 GPU with 8GB of VRAM. The implementation required Python 3.7 or later and PyTorch 1.12.1. The training took approximately 40 minutes.

IV. Results

A. Accuracy on the Real-World Data

To evaluate the LSTM model for classifying real EEG signals, we used the dataset consisting of 108 segmented files from the test subset, each containing 100 time units.

The classification report of the model (see Table I) demonstrates a high degree of accuracy and consistency of results, expressed through the metrics of precision, recall, and F1 score for each of the classes. In particular, the F1 score for various action classes range from 0.84 to 0.90, highlighting the ability of the model to adequately interpret temporal dependencies in the EEG data. The overall weighted accuracy of the model, expressed as the average F1 score, was 0.87.

TABLE I. Classification report of the LSTM model evaluated on the real-world EEG data

Class	Precision	Recall	F1 Score	Support
0 (elbow extension)	0.82	0.86	0.84	108
1 (elbow flexion)	0.91	0.83	0.87	108
2 (forearm pronation)	0.86	0.88	0.87	108
3 (forearm supination)	0.91	0.83	0.87	108
4 (hand close)	0.88	0.92	0.90	108
5 (hand open)	0.84	0.87	0.85	108
Accuracy	0.87 (Weighted Average F1 score)			

B. Accuracy on the Synthetic Data

To evaluate the performance of the LSTM model on synthetic EEG signals generated with EEG-GAN, the model was

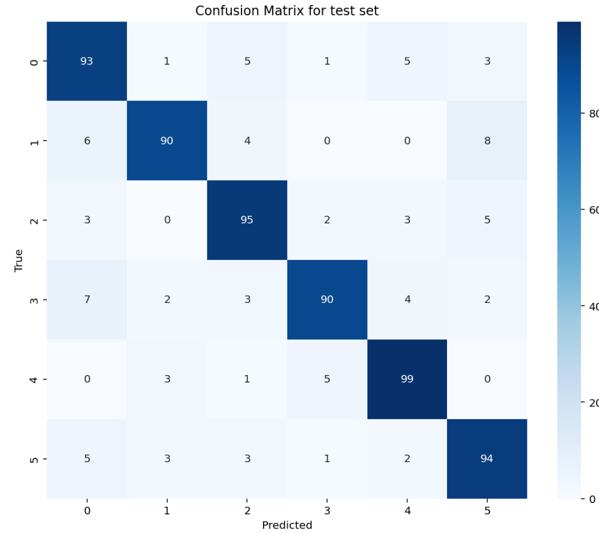

Fig. 4. Confusion matrix of the LSTM model evaluated on the real-world EEG data.

tested specifically on tasks involving opening and closing the hand. The model achieved an average F1 score of 0.76 on these tasks.

The classification report for the model (see Table II) indicates a high degree of precision with both classes achieving a score of 0.93. However, the recall scores were lower, at 0.72 for hand closing and 0.79 for hand opening, resulting in F1 scores of 0.81 and 0.86, respectively. These metrics indicate that while the model is rather precise in its predictions, its ability to capture all relevant instances in the synthetic dataset is somewhat limited.

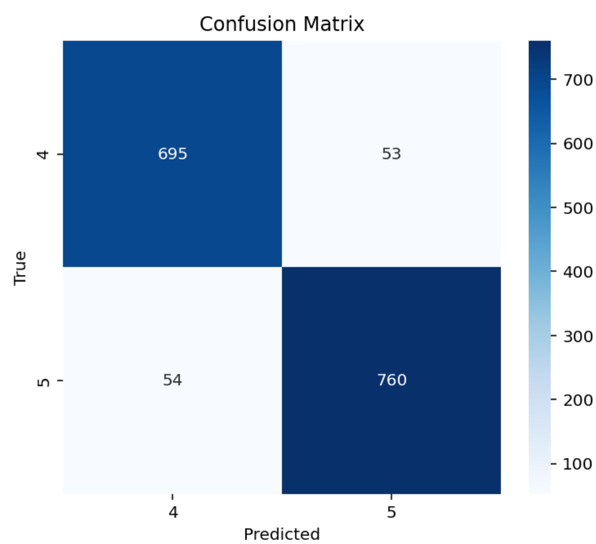

Fig. 5. Confusion matrix of the LSTM model evaluated on the synthetic EEG data.

TABLE II. Classification report of the LSTM model evaluated on the synthetic EEG data

Class	Precision	Recall	F1 Score	Support
4 (hand close)	0.93	0.72	0.81	960
5 (hand open)	0.93	0.79	0.86	960
Accuracy	0.76 (Weighted Average F1 score)			

V. Conclusion

In this study, we evaluated the effectiveness of LSTM neural networks for classifying EEG signals associated with various hand movements. The model trained on real-world data demonstrated high accuracy, achieving the average F1 score of 0.87. These results confirm the efficiency of LSTM networks in analyzing time series and their potential for developing robust brain-computer interfaces.

To assess the stability of the model against data variations and to investigate its generalization capabilities, we used synthetic data generated with EEG-GAN. These data were not used during training and served solely for testing. On this dataset, the average F1 score was 0.76, which is approximately 10 percentage points lower than when testing the model on the real-world data. Although there was a decrease in performance, the accuracy of the model remained well above the baseline (i.e, random guessing). This demonstrates a significant similarity between the artificially generated and the real-world data, underscoring the potential of synthetic data to augment small or scarce datasets.

Fig. 6 shows the first 100 data points for both real-world and synthetic signals of the T7 electrode. The visual inspection reveals that unique patterns are consistently reproduced in the synthetic data. This highlights the ability of EEG-GAN to replicate not only the general dynamics of EEG signals but also the unique characteristics of individual electrodes. In addition, the presence of synchronized peaks and other distinctive features across multiple electrodes indicates that the temporal relationships inherent to EEG signals are effectively captured.

Despite these promising outcomes, our findings also indicate that synthetic data generated with EEG-GAN do not fully replicate the dynamics of real-world EEG signals. This limitation is especially noticeable around 125 milliseconds in Fig. 6: while the real-world signal exhibits a smooth transition, the synthetic signal displays a stair-step pattern. Such discrepancies, although subtle, might influence classifier performance when synthetic data is used either exclusively or to augment real data.

One possible improvement that we intend to implement in future work is adjusting the segment length used for training the generator. Extending beyond the current 100 time units could provide the network with longer contextual information, thereby enhancing the quality and continuity of the generated EEG signals. Additionally, generating longer synthetic signals could help capture more complex and subtle EEG patterns necessary for accurate classification.

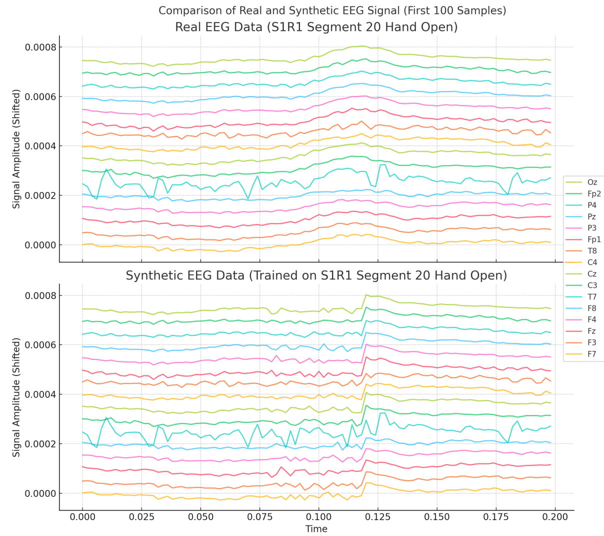

Fig. 6. Comparison of the real EEG signal from segment 20 and the corresponding synthetic signal generated during training. The plot shows the first 100 time points.

Another potential direction is to generate synthetic data for a broader range of motor imagery classes beyond just hand opening and closing. Until now, we focused on these two movements; however, creating synthetic EEG signals for other actions, such as elbow flexion, extension, or forearm pronation, could make the dataset more balanced, help the model learn better across all movement types, and support the development of more flexible and practical BCI systems.

In summary, our work confirms that LSTM networks are effective for EEG signal classification in the domain of motor imagery, and that synthetic data generated with GANs offer a promising way to expand data-scarce real-world datasets. The insights obtained from this study lay the groundwork for further development in EEG-based brain-computer interfaces, with future work focused on bridging the gap between synthetic and real data as well as on the integration of more sophisticated deep learning architectures.

Acknowledgment

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